MSAGLJS: Browsing Large Graphs with Tile Pyramids and Sleeve Routing

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Abstract

We present a new way to visualize a large graph in the style of online geographic maps. The method builds a *tile pyramid* for *semantic zoom*: at every zoom level the labels of the highest-ranked nodes remain readable, just as the names of major geographical features stay readable on those maps.

The edges are routed by a method we call *sleeve routing*, which computes for each edge the shortest path in its homotopy class around the node obstacles, using the Constrained Delaunay Triangulation. We apply several heuristics to speed up the routing.

We implemented our approach in the WebGL renderer of MSAGLJS, an open-source TypeScript library for graph visualization in web browsers, with the entire pipeline running client-side, without a dedicated server. We target graphs with at most 40K nodes and 100K edges. The renderer's demo can be seen at `https://microsoft.github.io/msagljs/renderer-webgl/index.html`.

MSAGLJS is available on GitHub[1] and as NPM packages[2].

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1 Introduction

Visualizing graphs in web browsers without a dedicated server presents unique challenges: the runtime is single-threaded, the per-tab JavaScript heap is capped at roughly 4 GB,[3] and users expect the smooth pan-and-zoom experience familiar from online maps. MSAGLJS is an open-source TypeScript library that addresses these challenges by combining efficient layout algorithms, a novel edge routing method, and a tiling scheme with semantic zoom.

MSAGLJS is consumed as a set of NPM packages and renders graphs using WebGL through the deck.gl framework [2]. It supports the layered Sugiyama scheme [44], Pivot MDS [14], and IPSep-CoLa [21], with edges routed around node obstacles. Throughout this paper our typical example is a social network laid out by IPSep-CoLa; the methods we describe, however, are agnostic to the layout algorithm and apply to any node placement in which the nodes do not overlap. The graph loading pipeline builds a hierarchy of tiles—analogous to map tiles—so that at any zoom level only a bounded number of entities are drawn. Tiles are delivered to the browser through the standard `TileLayer` of the `@deck.gl/geo-layers`

[1] `https://github.com/microsoft/msagljs`
[2] `@msagl/core`, `@msagl/drawing`, `@msagl/parser`, `@msagl/renderer-svg`, `@msagl/renderer-webgl` — all on `https://www.npmjs.com/`.
[3] In current Chrome, V8 with pointer compression caps the per-tab JavaScript heap at approximately 4 GB (2^{32} bytes).

package, giving smooth pan and zoom. Live demos are available online: a WebGL renderer for large graphs[4] and an SVG renderer for smaller, editable graphs[5].

This paper makes two main contributions:

1. A *tiling scheme* for large graph visualization that builds a hierarchical tile pyramid, limits visible entities per tile using PageRank-based filtering, simplifies edge routes at coarser levels, and integrates with the WebGL renderer for real-time pan and zoom.
2. A *sleeve routing* algorithm that routes edges on the dual graph of a Constrained Delaunay Triangulation, using per-source batched Dijkstra trees followed by the classical funnel algorithm to produce shortest-path geodesics in homotopy classes.

2 Related Work

2.0.0.1 Graph visualization tools.

Graphviz [25, 5] is a long-standing graph drawing tool supporting multiple layout algorithms, including the Sugiyama method and Scalable Force-Directed Placement [7]; it does not support tiling or WebGL rendering. Our library builds on the earlier MSAGL desktop layout engine [39] that did not have the features described here. On the web, Sigma.js [8] and Cytoscape.js [23] provide interactive graph rendering, but rely on straight-line or generic spline edges without obstacle-avoiding routing and without a pyramid of precomputed tiles. ReGraph [6] uses WebGL to render large graphs with straight-line edges and supports interactive cluster expansion but not tiling. Cosmograph [1] uses GPU-accelerated force-directed layout for million-node graphs with straight-line edges, without tiling. GraphMaps [38] introduced the tile-pyramid and node-ranking scheme on which we build, but it runs only on Windows, ships raster tiles, and reuses one global edge routing that is merely clipped per tile; MSAGLJS brings the idea to the browser with vector tiles and per-level sleeve rerouting. Cornac [41] handles huge graphs with tiles and edge bundling [34, 32] but requires a multi-server backend. Earlier navigation schemes for large graphs include ASK-GraphView [12], which organizes nodes into a hierarchical cluster tree, and topological fisheye views [24], which distort a multilevel layout [27, 33] under the focus; neither targets browser-side tile pyramids or per-level edge rerouting. Hierarchical edge bundling [31] and force-directed bundling [32] are complementary clutter-reduction techniques applied *after* edges have been routed.

2.0.0.2 Shortest paths and edge routing.

Our sleeve algorithm walks the dual of a Constrained Delaunay Triangulation [43] and relies on the classical funnel algorithm for extracting a geodesic from a triangle strip [36, 15, 26, 4]. The optimal algorithm for Euclidean shortest paths among polygonal obstacles [30] and the recent $O(m \log^{2/3} n)$ single-source shortest-path algorithm that breaks the sorting barrier [20] are remarkable theoretical results, but they are not practical: both build elaborate global data structures that we avoid. Graphviz builds the full visibility graph for edge routing in force-directed layouts, which has $O(n^2)$ edges. For Sugiyama layouts, it uses the funnel algorithm [18] but requires a simple containing polygon. yWorks [11] offers “Organic edge routing” based on force-directed simulation. The approach of Dwyer and Nachmanson [22] routes on a Yao-graph spanner with local shortcutting. For orthogonal layouts, the libavoid

[4] `https://microsoft.github.io/msagljs/renderer-webgl/index.html`
[5] `https://microsoft.github.io/msagljs/renderer-svg/index.html`

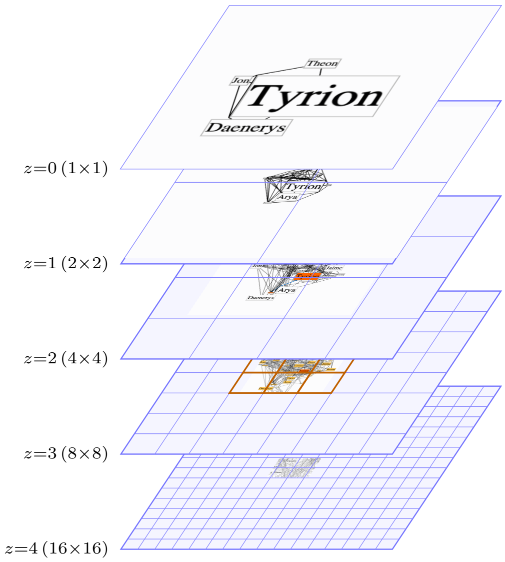


Figure 1 The entities rendered on each level as the user zooms in shown without scaling.

framework of Wybrow, Marriott and Stuckey performs obstacle-avoiding connector routing with incremental updates [45]. Our sleeve method eliminates the spanner entirely and routes directly on the CDT dual graph, and is, to our knowledge, the first routing algorithm integrated with a browser-side tile pyramid that reroutes edges per level.

2.0.0.3 CDT-based pathfinding.

The closest precedent to our routing pipeline is the CDT-and-funnel pathfinder of Demyen and Buro [17], originally designed for game-AI units of nonzero radius. They run admissible A* search on the CDT dual (with a reduced-graph variant that collapses corridor chains into single abstract edges) to recover provably shortest paths. Their search is, however, per-query: the bounds depend on the start and goal, so it cannot share work across the edges with the same source as our per-source Dijkstra does. For the graphs we target this would be roughly an order of magnitude slower; we therefore trade their proven optimality for speed.

3 Tiling

The input to tiling is a laid-out graph together with routed edges: we prefer to route the edges with sleeve routing (Section 4), to achieve the visual stability on a level change. The output is a sequence of $Z+1$ levels: level Z is the finest, and level 0 is the coarsest. Each level is a collection of tiles indexed by integer (x, y) grid coordinates.

A tile is a rectangular viewport together with its *tile data*: nodes, edge labels, edge arrowheads, and *edge clips*. An edge clip is a curve confined to the tile rectangle, paired with the non-empty list of graph edges whose routes enter and exit the tile at very close endpoints. Bundles usually capture the stretches on which several routes travel tight together inside a common sleeve; representing each such stretch as a single clip with an attached edge list, rather than as one clip per edge.

Pseudocode 1 describes the algorithm building the tile pyramid.

Algorithm 1 BuildTilePyramid($G, \mathcal{C}, \mathit{minTileSize}, \mathcal{M}_{\max}$)

$T_0 \leftarrow$ single tile holding all of G clipped to its bounding box
$\mathit{levels} \leftarrow [T_0]$; $z \leftarrow 0$; $\mathit{used} \leftarrow |T_0|$ ▷ $|T|$ = element count of tile T
while some tile in $\mathit{levels}[z]$ exceeds $\mathcal{C}$ elements **do**
    $\mathit{next} \leftarrow$ empty $2^{z+1} \times 2^{z+1}$ tile grid
    for all tiles $T \in \mathit{levels}[z]$ **do** ▷ per-tile inner loop
        split T into its four sub-tiles of next ▷ Section 3.2
        used += number of new elements added to next
        if $200 \cdot \mathit{used} > \mathcal{M}_{\max}$ **then** ▷ memory budget; checked mid-split
            discard next and **break** pyramid growth
        end if
    end for
    if next's tile width or height is below $\mathit{minTileSize}$ **then**
        break ▷ tile size below threshold
    end if
    $\mathit{levels}[z+1] \leftarrow \mathit{next}$; $z \leftarrow z + 1$
end while
$Z \leftarrow z$; $V_Z \leftarrow V(G)$; $s_v^{(Z)} \leftarrow 1$ for all $v \in V_Z$
$V_{\mathrm{PR}} \leftarrow$ nodes of G sorted by descending PageRank
for $z = Z - 1$ **down to** 0 **do**
    $(V_z, \{s_v^{(z)}\}) \leftarrow$ SelectTopKWithAdaptiveScale($V_{\mathrm{PR}}, \lceil |V|/2^{Z-z} \rceil$) ▷ Section 3.1
    build a CDT from the inflated polylines of V_z at scales $\{s_v^{(z)}\}$
    route every edge $e \in E_z$ on this CDT
    rebuild the tiles of $\mathit{levels}[z]$ ▷ Section 3.2
end for
return levels

3.0.0.1 How the index of the finest layer, Z, is chosen.

We treat the tiles differently when calculating Z and when preparing for rendering. There is no element filtering in the former case: all graph entities are represented in each level. We continue increasing Z, generating new levels, and split tiles, until a stopping condition is met. Then, while keeping the finest level intact, we rebuild the tiles of levels $0, \ldots, Z-1$ for rendering.

We start from $Z = 0$ and grow the pyramid one level at a time. We stop as soon as any of three conditions is met:

1. every tile at the new level contains at most $\mathcal{C}$ elements, with default $\mathcal{C} = 500$;
2. the new tile width and height are both below ten times the average node width and height;
3. while generating the tiles of the new finest level, the running total of stored tile elements summed over all levels, multiplied by 200 bytes per element, exceeds the memory budget $\mathcal{M}_{\max}$, with default 4 GB; the partial level is then discarded and Z is set to the previous finest level.

The third condition is checked incrementally inside the level-build loop, so we abort as soon as the memory barrier is reached. On every benchmark graph in Table 1 the first condition fires; the second and third are defensive guards that bound depth on hypothetical inputs whose finest level cannot be packed within $\mathcal{C} = 500$.

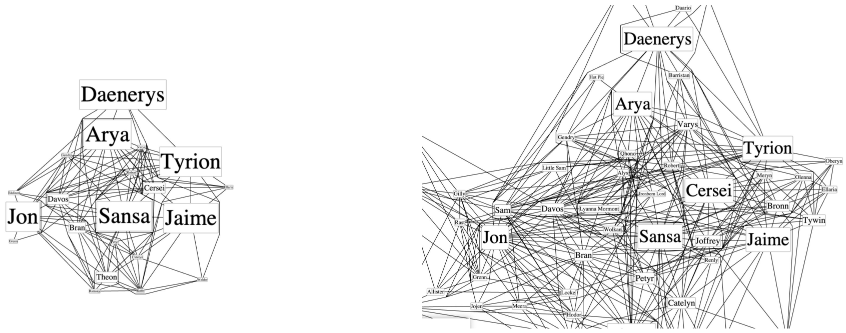


Figure 2 Two adjacent pyramid levels of the Game of Thrones graph rendered by the WebGL renderer of MSAGLJS with the sleeve router. The coarser level is on the left, the finer one on the right; both panels are cropped to the same world window. For each level a new graph is assembled from a PageRank-ranked prefix of the nodes, and edges are routed against the obstacles introduced at that level.

3.0.0.2 Three phases.

Once Z and the tile rectangles are fixed, each coarser level $z = Z-1, Z-2, \dots, 0$ is built in three phases:

1. pick the nodes of level z with PageRank and possibly enlarge them, Section 3.1;
2. reroute the edges of G_z around the resulting obstacles, Section 4;
3. recompute the per-tile edge clips on level z, Section 3.2.

Figure 2 shows two adjacent pyramid levels built by this procedure; Algorithm 1 summarizes the whole pipeline.

3.1 PageRank-Guided Level Construction

Before filling the level's tiles for rendering we run PageRank [40] on G once, and sort all nodes of the graph, V, by descending rank. Write $k = Z - z$ for the level's depth below the finest level. The candidates for level of depth k are the nodes of the *prefix* of the sorted array of length $\lceil |V|/2^k \rceil$. The level's edge set, used by Algorithm 1, is the subset of $E(G)$ whose endpoints both lie in this prefix.

Node positions are inherited from the original layout and never move: across levels we change only each node's display size.

We add nodes to a level of depth k by walking the level's prefix in rank-descending order. The top-ranked node is scaled by 2^k. We keep the node scales non-increasing, but as large as possible, while making sure that each next node does not overlap the previously selected nodes. If even at its original size the candidate's bounding box would overlap an already-accepted node's scaled box, the candidate is dropped.

Dropping a relatively high-ranked candidate at a coarse level is less disruptive than it sounds. Consider Arya's node in the *Game of Thrones* graph: as Figure 1 shows, it is dropped from the top layer because a higher-ranked node, Tyrion, has covered the available region. The user still sees that part of the graph as occupied, and Arya's individual node appears at a finer level where its box fits.

3.1.0.1 Spatial-hash overlap test.

A naive overlap test against every previously accepted node would dominate the cost of the filter. We instead maintain the accepted boxes in a uniform spatial hash whose cell size is

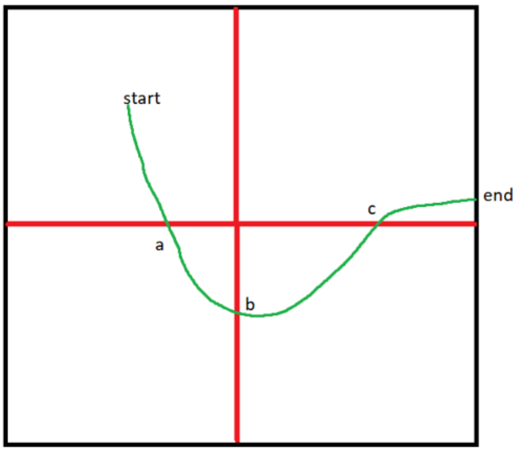


Figure 3 An edge clip with curve from *start* to *end* inside a tile. The two red midlines split the tile into four sub-tiles and meet the curve at a, b, c. Cutting the curve at these points yields four sub-clips $[start, a]$, $[a, b]$, $[b, c]$, $[c, end]$, each contained in one sub-tile and meeting that sub-tile's boundary only at its endpoints.

chosen so that any accepted box, and any candidate's box at the level's maximum scale, both fit inside one cell on each axis. Because of this choice, two such boxes can overlap only if their center cells differ by at most one step on each axis: testing a candidate therefore reduces to scanning the accepted boxes registered in the candidate's own cell and in the eight cells immediately around it.

The hash brings little benefit at the coarsest levels, where the prefix is short and even a naive scan over the few accepted boxes is cheap; it pays off at finer levels, where the prefix grows geometrically and a per-candidate constant-time test replaces a linear scan over hundreds of already-accepted boxes. On the layouts we tested the expected work per candidate is constant, so the overall cost is dominated by the initial sort by PageRank. In our experiments this phase was never a bottleneck: the per-level rerouting dominates the build time at every level we measured.

3.2 Splitting Edges between Tiles

We speed up edge-clip generation as follows. Let the *midlines* of a tile be the horizontal and vertical segments through its center; they split the tile into four sub-tiles. Consider a tile and an edge clip whose curve meets the tile boundary only at the curve's endpoints. Cutting the curve with the tile's midlines yields a set of sub-clips, each contained in one sub-tile, and each preserves the invariant: its curve meets the sub-tile boundary only at its endpoints. Because of this invariant, every split reuses the parent clip's two boundary crossings, and only the curve-vs-midline intersections need to be computed; no full curve-vs-rectangle clipping is required. Figure 3 illustrates one such split.

When we grow the pyramid in line 6 of Algorithm 1, each edge clip is already contained in its parent tile, so subdivision starts from that parent. When we prepare the tiles for rendering in line 23 of Algorithm 1, we have the rerouted graph G_z, so for each edge of G_z we start the subdivision from an initially empty temporary tile whose extent covers the whole graph.

Table 1 Maximum number of elements rendered in a single tile across the whole pyramid for several graphs (capacity $\mathcal{C} = 500$; Z chosen by the natural stops of Section 3). The capacity $\mathcal{C}$ only guides the choice of Z; we have no tight analytical bound on the actual rendered per-tile element count, which is therefore reported empirically. "Max tile's level" is the level on which the maximum tile was found. In the worst cases, `ca-HepPh` and `ca-CondMat`, the densest tile holds 6–8× the nominal capacity $\mathcal{C}$.

Graph	$\lvert V\rvert$	$\lvert E\rvert$	Levels built	Max per tile	Max tile's level
gameofthrones	407	2 639	5	324	4
composers	3 405	13 832	8	749	3
ca-GrQc	5 242	28 968	8	359	4
ca-HepTh	9 877	51 946	8	1 836	3
facebook_combined	4 039	88 234	9	928	5
ca-HepPh	12 008	236 978	9	3 643	3
ca-CondMat	23 133	186 878	8	2 949	3
deezer_europe	28 281	92 752	8	2 402	3
delaunay_n15	32 768	98 274	8	283	7

3.2.0.1 Bundling edge clips.

To correctly reflect the number of elements in a tile and further reduce the number elements in a tile, all clips within a tile that share the same, or almost the same, pair of endpoints are bundled into a single clip whose attached edge list accumulates every contributing edge; the bundled geometry is taken from the first such clip encountered during splitting, and the choice is consistent within each level. Table 1 reports the resulting maximum number of elements rendered in a single tile across the entire pyramid for a range of graphs.

4 Sleeve Routing on the CDT

To prepare for routing given the graph layout, each node is covered by a padded obstacle polygon. We compute a Constrained Delaunay Triangulation $\mathcal{T}$ on the set of these polygons [16, 19]. The *dual graph* D of $\mathcal{T}$ has one node per triangle and one edge for each pair of triangles sharing a side. The weight of an edge is the Euclidean distance between the centroids of its triangles.

4.1 Routing inside a Sleeve

We split routing an edge into two subproblems: a *combinatorial* one—which sequence of diagonals to cross—and a *geometric* one—which exact polyline through that sequence is shortest. The geometric step is the classical funnel algorithm [15, 29], which pulls the path taut inside the polygon formed by the sleeve triangles.

To obtain the sleeve we run a shortest-path search on D, where triangles inside obstacles other than the source and target obstacles are excluded. The natural per-edge instantiation is A* [28] with the straight-line distance to t as heuristic. When a node has many incident edges, however, running independent searches is wasteful. We instead group edges by source s and run a single Dijkstra computation per source, expanding until all targets are reached and recovering each sleeve by following parent pointers. This gives a speedup over A*; for example on Game of Thrones (407 nodes, 2 639 edges) it cuts routing time by ∼23 %. Table 3 reports the gain across the full benchmark suite.

We can improve even more. Since the query edges are undirected on D, each can be served from either endpoint, so the number of distinct roots is in turn reducible by reorienting query edges to minimize that count—an instance of minimum vertex cover on the demand graph, well approximated in linear time by the greedy maximum-degree rule and used as a source-side speed-up in many-to-many shortest-path frameworks [35]. Applying this heuristic reduces the number of Dijkstra trees from 331 (one per distinct source) to 198 on Game of Thrones and from 2 645 to 1 281 on the composers graph (3 405 nodes, 13 832 edges), shortening the routing pass by ∼28 % and ∼31 % respectively.

In addition, before launching the sleeve search for an edge we try routing the straight line segment between the source and the target by walking the CDT.

4.2 Edge Hint

We explored a routing variant in which the path generated for an edge on one level provides a hint for routing the same edge on an adjacent level, increasing the visual stability of the routes between levels. Its shortcoming is that the search cannot be batched per source as the Dijkstra-tree variant of Section 4.1 is, which impacts performance.

4.3 Vertex Collapse at Source and Target

We improve the sleeve's geometry in the situation depicted in Figure 4. The remedy is to *collapse* corner-hugging chain vertices to the corresponding endpoint. Walking the right chain of the sleeve outward from s, we find the first vertex of the source obstacle at which the chain turns right toward s, and replace every source-owned vertex on that chain up to and including this one by s. The left chain is processed mirror-symmetrically (with left turns instead of right), and the target end likewise. Diagonals whose two endpoints coincide after collapse are dropped, and the left/right orientation of each surviving diagonal is preserved from the uncollapsed geometry. The collapsed vertex in Figure 4(a) is marked in yellow.

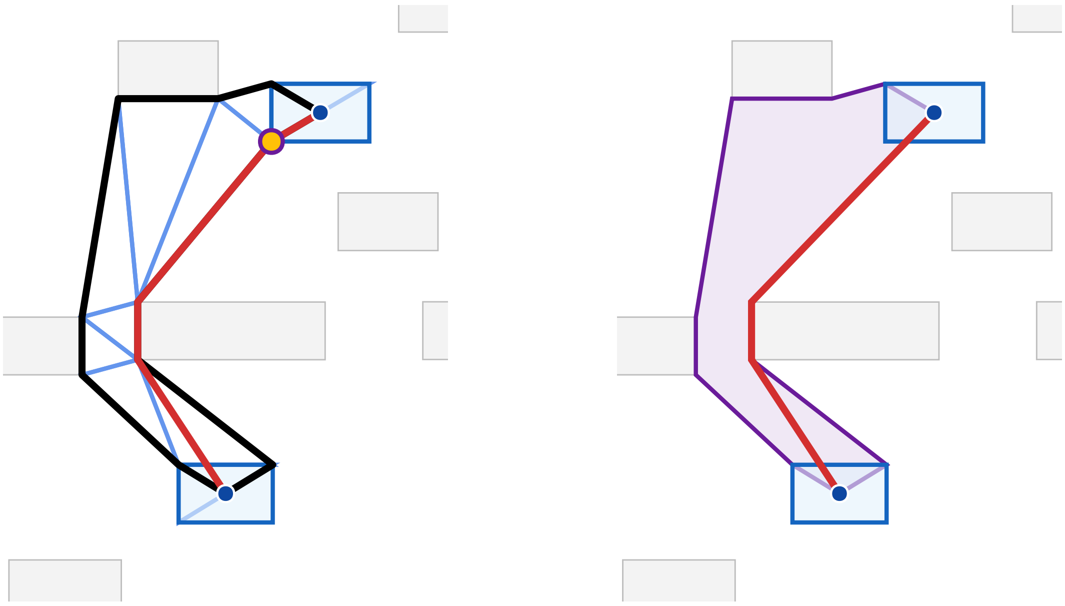
After collapse, the funnel sees a wide opening at each endpoint, as in Figure 4(b), and naturally selects the optimal exit direction. A final arrowhead-trimming step clips the path at the node boundary curves. Collapse might cause the resulting path to overlap nodes other than the source and the target, but we have found the effect to be negligible.

The implementation also offers an optional pass that fits a Bezier segment into each polyline corner to render edges as smooth curves. It is disabled by default for performance: the polyline rendering is fast and visually acceptable, so we trade the smoother curves for routing speed.

5 Rendering with deck.gl

The output of the build pipeline of Section 3 is consumed in the browser by the `TileLayer` of `@deck.gl/geo-layers` [10]. `TileLayer` is the standard component used by online tile-map renderers: it tracks the viewport, determines which tiles are needed at the current zoom, and asynchronously requests them through a user-supplied `getTileData` callback, caching the results across frames so that subsequent pans within the cache require no further work.

MSAGLJS integrates with this component at construction time by exposing the precomputed pyramid as the `TileLayer`'s zoom range, with the coarsest pyramid level mapped to the layer's lowest zoom and the finest level to the highest. As the user pans and zooms, `TileLayer` resolves the viewport to an integer zoom and a set of axis-aligned tile indices and, for each tile it needs to display, calls `getTileData`. The renderer translates each such



(a) before collapse (b) after collapse

Figure 4 Effect of vertex collapse on the LORAS–RENLY edge of the Game of Thrones graph. Endpoints are shown as blue dots; their padded obstacles are highlighted in light blue. (a) The CDT triangles forming the sleeve are drawn in cornflower blue and the funnel path in red; the thick black polyline is the perimeter of the non-collapsed sleeve. The yellow dot marks the source-obstacle vertex that gets collapsed. (b) The sleeve after collapse, drawn as a single purple polygon with the source/target obstacles merged into the corresponding endpoints, and the funnel path (red) through it. Collapsing widens the funnel opening at the endpoint and removes the spurious detour around the obstacle corner.

request back into a tile of the corresponding pyramid level and returns its precomputed nodes, edge clips, edge labels, and arrowheads. The mapping is one-to-one, so no geometric work is performed at view time: panning shifts the visible window over the tile cache, and zooming displays tiles from the adjacent level using `TileLayer`'s built-in cross-fade.

The rendered geometry depends only on the zoom level and does not change with the viewport, except for the clipping to the visible window. Within each level, the displayed nodes are the ones chosen for that level by the construction of Section 3.1, and the displayed edges are the geodesics around those nodes, computed by the sleeve router of Section 4.

5.0.0.1 Interaction.

The renderer supports pan, zoom, and hover highlighting of nodes and edges. Hovering an edge highlights it and its incident nodes; hovering a node highlights the node, its one-hop neighbors, and its two-hop neighbors, each in a different color. A `zoomTo` API animates the viewport to a target rectangle with a smooth transition and is used, for example, by the search box to focus the view on a query result.

6 Experimental Results

We evaluate the sleeve router and the tile-pyramid build on the same benchmark graphs as Table 1: the Game of Thrones social network [13], the GD 2011 contest `composers` graph [9],

Graph	$\lvert V \rvert$	$\lvert E \rvert$	Routing	Tiling	Total
gameofthrones	407	2 639	0.07	0.07	0.38
composers	3 405	13 832	1.49	0.74	2.69
ca-GrQc	5 242	28 968	1.78	0.87	3.40
facebook_combined	4 039	88 234	4.34	2.36	7.77
ca-HepTh	9 877	51 946	9.33	3.41	14.87
ca-HepPh	12 008	236 978	38.89	14.08	57.71
ca-CondMat	23 133	186 878	79.72	20.52	110.06
deezer_europe	28 281	92 752	134.78	23.09	172.68
delaunay_n15	32 768	98 274	18.71	5.64	41.38

Table 2 End-to-end loading benchmarks in headless Chrome 148 on an Apple M4 Max laptop, all times in seconds. Measurements come from the `chrome-routing-bench` example, which runs the unmodified browser bundle of MSAGLJS. *Routing* sums the CDT-construction and Dijkstra-tree sleeve-search phases of Section 4.1 on the full graph; *Tiling* is the build of the tile pyramid, capacity $\mathcal{C} = 500$ with Z chosen by the natural stops of Section 3, which includes the per-level rerouting; *Total* is the total loading time including parsing of the graph file, graph layout, and all other stages.

four SNAP collaboration networks (`ca-GrQc`, `ca-HepTh`, `ca-HepPh`, `ca-CondMat`) and the SNAP `facebook_combined` ego network [37, 3], the `deezer_europe` social network [42], and the SuiteSparse Delaunay mesh `delaunay_n15`. All experiments run in headless Chrome 148 on an Apple M4 Max laptop with 48 GB of unified memory, with the unmodified browser bundle of MSAGLJS loaded into the page. Layout uses IPsepCola, and the tile pyramid is built with the same parameters as Table 1: capacity $\mathcal{C} = 500$, with Z chosen by the natural stops of Section 3.

Table 2 reports per-graph timings of three pipeline stages: *Routing*, the sum of the two phases logged by the sleeve router on the full graph—constrained Delaunay triangulation of the obstacles and the Dijkstra-tree search on the CDT dual (Section 4.1); *Tiling*, the build of the tile pyramid (Section 3), which includes the per-level rerouting; and *Total*, the end-to-end loading time including parsing.

The CDT phase contributes negligibly to the routing column—from 6 ms on Game of Thrones to about half a second on the largest graphs—so virtually all of *Routing* is the Dijkstra-tree sleeve search, with the vertex-cover heuristic of Section 4.1 reducing the number of trees we run. Even on graphs with hundreds of thousands of edges (`ca-CondMat`, `ca-HepPh`, `deezer_europe`) the full pipeline finishes in a few minutes; this is a one-time build cost. The resulting tile pyramid already contains the per-level routes, so panning and zooming only fetch and render the precomputed tiles, with no rerouting at view time.

Table 3 isolates the contribution of the two source-side optimizations of Section 4.1: batched Dijkstra trees over per-edge A* and the vertex-cover heuristic on top of that. Each row times a single sleeve-search pass run after layout, with the constrained Delaunay triangulation excluded so that only the search costs are compared. The batched Dijkstra trees beat per-edge A* on every social graph, with the gain growing with the average degree and reaching +63 % on `facebook_combined` and +48 % on `ca-HepPh`; the vertex-cover heuristic then shaves a further 11–42 % off those same graphs. The Delaunay mesh `delaunay_n15` is the only outlier: its source set is almost in bijection with its vertices, so neither optimization has many opportunities to share paths and the bookkeeping leaves them marginally slower than the baselines.

			Time (s)			Speedup	
Graph	Srcs	VC	A*	DJ	VC_DJ	DJ/A*	VC/DJ
gameofthrones	331	198	0.077	0.059	0.043	+23 %	+28 %
composers	2 645	1 281	2.60	2.14	1.49	+18 %	+31 %
ca-GrQc	5 241	2 796	2.55	2.52	1.65	+1 %	+35 %
facebook_combined	3 663	3 045	13.31	4.88	4.36	+63 %	+11 %
ca-HepTh	9 875	5 003	17.13	15.42	9.28	+10 %	+40 %
ca-HepPh	12 006	7 032	111.2	58.07	36.38	+48 %	+37 %
ca-CondMat	23 133	13 564	189.1	131.1	76.45	+31 %	+42 %
deezer_europe	21 060	13 450	172.1	154.0	124.4	+11 %	+19 %
delaunay_n15	28 467	23 456	13.57	16.02	14.88	−18 %	+7 %

Table 3 Sleeve-search times of the three routing modes on the benchmark graphs of Table 2, in headless Chrome 148. Columns: *Srcs*, the number of distinct edge endpoints in the demand graph—one per source for the per-edge A* baseline; *VC*, the number of Dijkstra trees actually launched after the greedy maximum-degree vertex-cover reorientation of Section 4.1; *A**, wall-clock seconds of one full sleeve-search pass with per-edge A* (straight-line distance to the target as heuristic); *DJ*, wall-clock seconds with the batched Dijkstra-tree mode—one tree per source, expanding until all of that source's targets are reached; *VC_DJ*, wall-clock seconds with the vertex-cover-reoriented Dijkstra-tree mode—same as DJ but with only *VC* roots; *DJ/A** = (A* − DJ)/A*, speedup of DJ over A*; *VC/DJ* = (DJ − VC_DJ)/DJ, additional speedup of VC_DJ over DJ. The CDT-construction phase, identical across modes, is excluded so that only the search costs are compared.

7 Conclusion

We presented MSAGLJS, an open-source TypeScript library for graph layout, edge routing, and visualization in web browsers. The sleeve routing method routes edges directly on the CDT dual graph using batched Dijkstra trees and the funnel algorithm. The tiling scheme enables map-like exploration of large graphs with level-of-detail filtering and edge simplification. The WebGL renderer, built on deck.gl, provides smooth pan-and-zoom interaction.

7.0.0.1 Limitations.

The capacity $\mathcal{C}$ guides the choice of Z but does not bound the post-rebuild per-tile element count: coarser levels are reassembled from PageRank-filtered candidates once Z is fixed, so a rebuilt level holds fewer edges than the finest level but spreads them over correspondingly fewer tiles. Table 1 shows that on the densest collaboration graphs, `ca-HepPh` and `ca-CondMat`, the heaviest rebuilt tile holds 6–8× $\mathcal{C}$, and on every benchmark graph except Game of Thrones and `delaunay_n15` the peak falls at an intermediate level rather than the finest. Browser memory budgeting therefore needs a per-level safety margin, and tiles with extreme density may need additional client-side culling at render time.

7.0.0.2 Future work.

Three directions look promising. First, the capacity check can be extended to the rebuilt levels by treating the post-filter element count as a refinement target and locally splitting any over-capacity tile into a sub-grid, so that the global Z does not have to grow. Second, head-to-head measurements against Cornac, GraphMaps, Cosmograph, and ReGraph on a shared corpus would let users compare load time, memory, and routing quality directly; the present paper reports only MSAGLJS's own pipeline, since each of those tools differs in input format, output features, and deployment model. Third, the tile size is currently fixed in screen pixels; tuning it for high-DPI displays and multi-monitor setups, and adapting it to mobile viewports, would extend the renderer to a broader range of clients.

MSAGLJS is released under the MIT license.

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